

PRIMEOR SOLUTIONS

E-Commerce Data Analytics Internship

LEVEL 2 – SQL DATA ANALYSIS

FINAL BUSINESS INSIGHTS REPORT

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E-COMMERCE DATA ANALYTICS INTERNSHIP

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1. PROJECT OVERVIEW

This report presents the SQL analysis of the cleaned e-commerce dataset provided for the internship. The analysis focuses on product profitability, customer sales, regional and market performance, category profitability, discounting, loss-making records, monthly sales trends, sub-categories, shipping modes, and additional business KPIs.

Dataset size: 51,271 records

Unique orders: 25,029

Total sales: 12,635,222.00

Total profit: 1,467,497.61

Average discount: 14.29%

Negative-profit records: 12,538

2. OBJECTIVES

- Identify the most profitable products and valuable customers.
- Compare sales and profit across regions and markets.
- Evaluate category and sub-category performance.
- Identify discounting patterns and loss-making records.
- Analyse monthly and yearly sales trends.
- Understand shipping-mode usage.
- Generate actionable business insights using SQL.

3. KEY BUSINESS FINDINGS

3.1 Overall Performance

The dataset contains 51,271 records and 25,029 unique orders.

Total sales are 12,635,222.00, while total profit is 1,467,497.61.

The overall average discount is 14.29%.

3.2 Top 10 Profitable Products

The highest-profit product is "Canon imageCLASS 2200 Advanced Copier" with profit of 25,199.93.

The remaining leading products are:

- Canon imageCLASS 2200 Advanced Copier: 25,199.93
- Cisco Smart Phone, Full Size: 17,238.52
- Motorola Smart Phone, Full Size: 17,027.11
- Hoover Stove, Red: 11,807.97
- Sauder Classic Bookcase, Traditional: 10,672.07

- Harbour Creations Executive Leather Armchair, Adjustable: 10,427.33
- Nokia Smart Phone, Full Size: 9,938.20
- Cisco Smart Phone, with Caller ID: 9,786.64
- Nokia Smart Phone, with Caller ID: 9,465.33
- Belkin Router, USB: 8,955.02

3.3 Top 10 Customers by Sales

The highest-sales customer is "Tom Ashbrook" with sales of 40,489.00.

Top customers:

- Tom Ashbrook: 40,489.00
- Tamara Chand: 37,453.00
- Greg Tran: 35,552.00
- Christopher Conant: 35,187.00
- Sean Miller: 35,131.00
- Bart Watters: 32,315.00
- Natalie Fritzler: 31,778.00
- Fred Hopkins: 30,404.00
- Jane Waco: 30,288.00
- Hunter Lopez: 30,246.00

3.4 Region-wise Sales

The highest-sales region is Central with sales of 2,821,740.00.

Its total profit is 311,513.19.

The lowest-sales region in this dataset is Canada with sales of 66,932.00.

3.5 Category Analysis

The category with the highest average profit per record is Technology at 65.46.

The category with the highest total profit is Technology with 663,200.15.

The highest average discount is observed in Furniture at 16.80%.

3.6 Market-wise Revenue

The highest-revenue market is APAC with revenue of 3,581,785.00.

Its total profit is 436,991.31.

The next highest-revenue market is EU with revenue of 2,937,166.00.

4. LOSS-MAKING ANALYSIS

There are 12,538 records with negative profit.

This indicates a significant opportunity to investigate discounting, product-level margins, shipping costs, and other factors associated with loss-making transactions.

The SQL query filters negative-profit records so that specific products and orders can be reviewed.

5. SUB-CATEGORY PERFORMANCE

The most profitable sub-category is Copiers with total profit of 258,174.35.

The highest-sales sub-category is Phones with sales of 1,706,642.00.

6. SHIPPING MODE ANALYSIS

The most commonly used shipping mode by unique orders is Standard Class with 15,151 orders.

Shipping-mode analysis can help evaluate operational demand and customer delivery preferences.

7. YEAR-WISE PERFORMANCE

- 2011: Sales = 2,259,569.00, Profit = 248,672.95, Orders = 4,440
- 2012: Sales = 2,677,479.00, Profit = 307,411.88, Orders = 5,343
- 2013: Sales = 3,404,254.00, Profit = 408,261.44, Orders = 6,720
- 2014: Sales = 4,293,920.00, Profit = 503,151.35, Orders = 8,526

The dataset shows growth in total sales from 2,259,569.00 in 2011 to 4,293,920.00 in 2014.

8. ADVANCED SQL ANALYSIS INCLUDED

- Profit margin by category
- Top products by profit margin
- Year-wise sales and profit
- Segment-wise performance
- High-discount and loss-making records
- Region + category performance
- Average order value
- Customer purchase frequency
- Most profitable region

9. RECOMMENDATIONS

1. Focus on high-profit products and sub-categories when planning inventory and promotions.
2. Review high-discount transactions that generate negative profit.
3. Investigate the large number of loss-making records at product and order level.
4. Maintain strong customer relationships with the highest-sales customers.
5. Use market and regional performance to prioritize sales and marketing resources.
6. Monitor yearly and monthly sales trends to support demand planning.
7. Review shipping-mode usage to balance customer expectations and logistics costs.

10. CONCLUSION

SQL analysis converts the cleaned e-commerce dataset into measurable business insights.

The analysis identifies profitable products and customers, compares markets and regions, highlights discount and loss patterns, and evaluates sales trends and shipping usage.

The advanced queries extend the mandatory Level 2 requirements and provide a stronger foundation for the Power BI dashboard in the next stage.

FILES INCLUDED

- Level_2_SQL_Queries.sql – complete SQL queries
- Level_2_Final_Business_Insights_Report.pdf – final report